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5.Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

TASK1:

CODE:

```
lasya_grandhi@DESKTOP-7Q7  ×  +  ▾
GNU nano 8.7.1 Practic
    clock_gettime(CLOCK_MONOTONIC, &end);

    double time_taken =
        (end.tv_sec - start.tv_sec) +
        (end.tv_nsec - start.tv_nsec) / 1e9;

    printf("\nProducer: Sent %d values\n", n);
    printf("Communication time: %.6f seconds\n", time_taken);
    printf("Communication rate: %.2f values/second\n",
        n / time_taken);
}
else
{
    close(pipefd[1]);
    int count = 0;

    while (read(pipefd[0], &data, sizeof(data)) > 0)
    {
        count++;
    }

    close(pipefd[0]);

    printf("Consumer: Received %d values\n", count);
    exit(0);
}
return 0;
}
```

OUTPUT:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano Practical5.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc Practical5.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
Consumer: Received 100000 values

Producer: Sent 100000 values
Communication time: 0.028403 seconds
Communication rate: 3520753.24 values/second
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```

TASK2:

CODE:

```
GNU nano 8.7.1 Practica
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main(){
    int pipefd[2];
    pid_t pid1, pid2;
    if (pipe(pipefd) == -1){
        perror("pipe");
        exit(1);
    }
    pid1 = fork();
    if (pid1 < 0){
        perror("fork");
        exit(1);
    }
    if (pid1 == 0)
    {
        close(pipefd[0]);
        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[1]);
        execlp("ls", "ls", "-l", NULL);
        perror("execlp ls");
        exit(1);
    }
    pid2 = fork();
    if (pid2 < 0){
        perror("fork");
        exit(1);
    }
}
```

```
        exit(1);
    }
    pid2 = fork();
    if (pid2 < 0){
        perror("fork");
        exit(1);
    }

    if (pid2 == 0)
    {
        close(pipefd[1]);
        dup2(pipefd[0], STDIN_FILENO);
        close(pipefd[0]);
        execlp("grep", "grep", ".c", NULL);
        perror("execlp grep");
        exit(1);
    }
    close(pipefd[0]);
    close(pipefd[1]);
    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);
    printf("Command completed.\n");
    return 0;
}
```

OUTPUT:

```
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ nano Practical5b.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ gcc Practical5b.c
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ ./a.out
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 744 Sep  3 05:11 Practical3.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 422 Sep 10 04:49 Practical4.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 300 Sep 10 04:58 Practical4b.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 1290 Sep 10 05:28 Practical5.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 956 Sep 10 05:36 Practical5b.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 159 Jul 27 08:32 addition.c
-rwxr-xr-x 1 lasya_grandhi lasya_grandhi 16104 Jul 28 04:57 arithmetic
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 540 Jul 28 04:56 arithmetic.c
-rwxr-xr-x 1 lasya_grandhi lasya_grandhi 16112 Jul 28 05:51 arithmetic_switch
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 1161 Jul 28 05:50 arithmetic_switch.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 89 Aug 24 06:01 c2
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 155 Aug  3 07:10 execs.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 343 Aug  3 14:52 execworking.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 184 Aug  4 06:01 file.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 339 Aug 25 05:50 open.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 174 Aug  1 09:51 p2.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 533 Aug  3 06:53 p2.c.save
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 322 Aug 24 05:50 practical2.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 0 Jul 30 06:03 process1
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 150 Jul 30 06:01 process1.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 215 Aug  1 09:42 process2.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 238 Aug 11 04:47 program1.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 630 Aug 24 05:47 question1.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 633 Aug 18 03:17 question1.c.save
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 607 Aug  4 05:20 question2.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 1023 Aug 11 05:54 question3.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 341 Aug 11 05:57 question4.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 1050 Aug 12 17:19 question5.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 662 Aug 12 17:21 question6.c
-rw-r--r-- 1 lasya_grandhi lasya_grandhi 325 Aug 25 06:09 read.c
Command completed.
lasya_grandhi@DESKTOP-7Q7B8KJ:~$ |
```